Chapter 24 Test

Name:_____ Student ID Number: _____ Class No. _____

I. Single Choices (80′)

1	2	3	4	5	6	7	8	9	10
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**		

1. The charge on the square plates of a parallel-plate capacitor is Q. The potential across the plates is maintained with constant voltage by a battery as they are pulled apart to twice their original separation, which is small compared to the dimensions of the plates. The amount of charge on the plates is now equal to ( )

   A) $4Q$. B) $2Q$. C) Q. D) $Q/2$.

2. The electric field between square the plates of a parallel-plate capacitor has magnitude E. The potential across the plates is maintained with constant voltage by a battery as they are pulled apart to twice their original separation, which is small compared to the dimensions of the plates. The magnitude of the electric field between the plates is now equal to ( )

   A) $4E$. B) $2E$. C) E. D) $E/2$.

3. Equal but opposite charges Q are placed on the square plates of an air-filled parallel-plate capacitor. The plates are then pulled apart to twice their original separation, which is small compared to the dimensions of the plates. Which of the following statements about this capacitor are true? (There may be more than one correct choice.) ( )

   A) The energy density in the capacitor has increased. B) The electric field between the plates has increased.

   C) The potential difference across the plates has doubled. D) The capacitance has doubled.

4. When two or more capacitors are connected in series across a potential difference ( )

   A) the potential difference across the combination is the algebraic sum of the potential differences across the individual capacitors.

   B) each capacitor carries the same amount of charge.

   C) the equivalent capacitance of the combination is less than the capacitance of any of the capacitors.

   D) All of the above choices are correct.

5. When two or more capacitors are connected in parallel across a potential difference ( )

   A) the potential difference across each capacitor is the same.

   B) each capacitor carries the same amount of charge.

   C) the equivalent capacitance of the combination is less than the capacitance of any of the capacitors.

   D) All of the above choices are correct.

6. Fig. 1 shows 3 capacitors, of equal capacitance C, connected to a battery of voltage V. What is the equivalent capacitance of this combination? ( )

   A) 3C/2 B) C/2

   C) 2C/3 D) 3 C

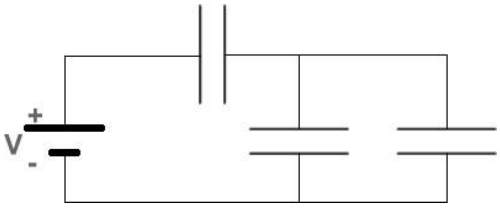


Fig. 1

7. What charge appears on the plates of a $6.0-\mu F$ capacitor when it is charged to $40\,V$? ( )

   A) $60\,\mu C$ B) $120\,\mu C$ C) $180\,\mu C$ D) $240\,\mu C$

8. The magnitude of the charge on each plate of a parallel plate capacitor is $4\ \mu C$ and the potential difference between the plates is 80 V. What is the capacitance of this capacitor? ( )

A) $300 \times 10^{-6}\ F$ B) $100 \times 10^{-6}\ F$ C) $20 \times 10^{-6}\ F$ D) $5 \times 10^{-8}\ F$

9. A parallel-plate capacitor has plates of area $0.80\ m^2$ separated by a distance of $3.0\ mm$. What is this capacitor's capacitance? ( )

A) $270\ F$ B) $70\ F$ C) $2.4 \times 10^{-9}\ F$ D) $4.8 \times 10^{-10}\ F$

10. A parallel-plate capacitor has a voltage $V = 6.0\ V$ between its plates. Each plate carries a surface charge density $\sigma = 7.0\ nC/m^2$. What is the separation of the plates? ( )

A) $2.5\ mm$ B) $5.6\ mm$ C) $7.6\ mm$ D) $8.2\ mm$

11. A parallel-plate capacitor has a plate separation of $5.0\ cm$. If the potential difference between the plates is $2000\ V$, with the top plate at the higher potential, what is the electric field between the plates? ( )

A) $400\ N/C$ upward B) $400\ N/C$ downward

C) $40000\ N/C$ upward D) $40000\ N/C$ downward

12. Three capacitors are connected as shown in Fig. 2. What is the equivalent capacitance between points A and B? ( )

A) $1.7\ \mu F$ B) $4.0\ \mu C$

C) $7.1\ \mu F$ D) $12\ \mu F$

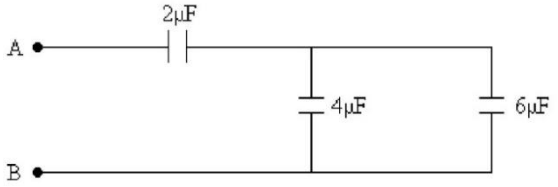


Fig. 2

13. What is the equivalent capacitance of the capacitor network shown in Fig. 3? (All the capacitors have a capacitance $C = 5\ \mu F$. The voltage of the battery is $V = 6V$.) ( )

A) $20\ \mu F$ B) $3\ \mu F$

C) $10\ \mu F$ D) $5\ \mu F$

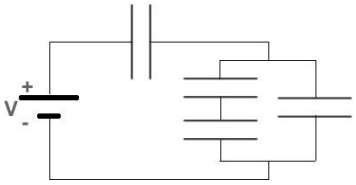


Fig. 3

14. A $1\ pF$ capacitor is connected in parallel with a $2\ pF$ capacitor, the parallel combination then being connected in series with a $3\ pF$ capacitor. The resulting equivalent capacitance is ( )

A) $1.5\ pF$. B) $6\ pF$. C) $3\ pF$. D) $4.5\ pF$.

15. A $4 - \mu F$ capacitor has a potential drop of $20\ V$ between its plates. The electric potential energy stored in this capacitor is ( )

A) $0.8\ \mu J$ B) $8\ \mu J$ C) $80\ \mu J$ D) $800\ \mu J$

16. A parallel plate capacitor has a potential difference between the plates of $80\ V$. If the charge on one of the plates of the capacitor is $+8.0\ \mu C$, what is the electrical energy stored by this capacitor? ( )

A) $640 \times 10^{-6}\ J$ B) $320 \times 10^{-6}\ J$ C) $5.0 \times 10^{-8}\ J$ D) $6.0 \times 10^{-8}\ J$

17. Two large non-conducting plates of surface area $A = 0.25\ m^2$ carry equal but opposite charges $Q = 75\ \mu C$. What is the energy density of the electric field between the two plates? ( )

A) $1.7 \times 10^{7}\ J/m^3$ B) $5.1 \times 10^{3}\ J/m^3$ C) $3.4 \times 10^{7}\ J/m^3$ D) $2.5 \times 10^{3}\ J/m^3$

18. A parallel-plate capacitor stores a charge $Q = 4.00\ nC$ when connected to a battery of voltage $V = 10.0\ V$. The energy density is then $u = 3.62 \times 10^{-4}\ J/m^3$. What is the surface area of the plates? ( )

A) $0.0250\ m^2$ B) $0.0500\ m^2$ C) $0.0923\ m^2$ D) $0.1030\ m^2$